

JUN YAMAMOTO

PHD CANDIDATE IN NETWORK SCIENCE

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RESEARCH PROFILE

I investigate the organizing principles underlying network structure and dynamics using statistical physics. My current work focuses on physical networks, network coarse-graining and scale invariance, and network diffusion and synchronization.

EDUCATION

- 2023–present PhD in Network Science, Central European University, Vienna, Austria.
Supervisors: Prof. Dr. Márton Pósfai and Prof. Dr. János Kertész
Thesis: “Dynamical processes on physical networks” (tentative).
- Jun–Jul 2026 Visiting Student, Department of Physics, Northeastern University, Boston, USA.
Host PI: Prof. Dr. Albert-László Barabási
- 2022–2023 MSc in Mathematics, Queen Mary University of London, London, UK.
Supervisor: Prof. Dr. Ginestra Bianconi
Thesis: “Modelling higher-order network dynamics in the presence of triadic interactions.”
- 2019–2020 Exchange Student, Department of Physics, ETH Zürich, Zürich, Switzerland.
- 2017–2022 BEng in Applied Physics, Hokkaido University, Sapporo, Japan.
Supervisor: Prof. Dr. Kousuke Yakubo
Thesis: “Bifractality of fractal scale-free networks.”

PEER-REVIEWED JOURNAL ARTICLES

- [3] M. Niedostatek, A. Baptista, **J. Yamamoto**, J. Kurths, R. Sanchez Garcia, B. MacArthur, and G. Bianconi. “Mining higher-order triadic interactions.” *Nature Communications* **16**, 11613 (2025). DOI: 10.1038/s41467-025-66577-z
- [2] K. Yakubo, G. Shimojo, and **J. Yamamoto**. “Random walks on bifractal networks.” *Physical Review E* **110**, 064318 (2024). DOI: 10.1103/PhysRevE.110.064318
- [1] **J. Yamamoto** and K. Yakubo. “Bifractality of fractal scale-free networks.” *Physical Review E* **108**, 024302 (2023). DOI: 10.1103/PhysRevE.108.024302

PREPRINTS/UNDER REVIEW

- [3] S. Mizutaka, **J. Yamamoto**, and K. Yakubo. “Structural crossover of complex networks: bridging degree correlation and fractality.” arXiv:2610.02883 (2026).
- [2] **J. Yamamoto**, I. Bonamassa, and M. Pósfai. “Localization transitions of diffusion dynamics in physical networks.” arXiv:2607.19486 (2026). Under review at *Physical Review Letters*.
- [1] **J. Yamamoto**, I. Bonamassa, and M. Pósfai. “Laplacian eigenmode localization in physical complex networks.” Under review at *Physical Review E* (2026). Joint submission with arXiv:2607.19486.

WORKING PAPERS

- J. Yamamoto**, C. Geier, and M. Pósfai. “Synchronization in heterogeneous modular networks.” *Manuscript in preparation*.
- J. Yamamoto**, M. Pósfai, G. Bianconi, A.-L. Barabási, et al. “Synchronization in spin Hall nano-oscillator networks.” With J. Åkerman’s group, University of Gothenburg.

CONFERENCE PRESENTATIONS

† Presenter

- 2026 S. Mizutaka, **J. Yamamoto**[†], and K. Yakubo. “Long-range degree correlations reveal structural crossover in critical random graphs.” Complexity of Life Conference (CoL2026), Graz, Austria. September 7, 2026.
- 2025 **J. Yamamoto**[†], I. Bonamassa, and M. Pósfai. “Localization-delocalization transition in physical networks.” The 29th International Conference on Statistical Physics (StatPhys29), Florence, Italy. July 14, 2025.
- 2025 **J. Yamamoto**[†], I. Bonamassa, and M. Pósfai. “Delocalization of physical Laplacian eigenvectors due to degree-volume correlation.” The International School and Conference on Network Science (NetSci2025), Maastricht, the Netherlands. June 4, 2025.
- 2022 **J. Yamamoto**[†] and K. Yakubo. “Bifractal property of stochastic scale-free networks.” The Physical Society of Japan 77th Annual Meeting, online. March 15, 2022.
- 2021 **J. Yamamoto**[†] and K. Yakubo. “Multifractality of scale-free networks.” The Physical Society of Japan 76th Annual Meeting, online. September 20, 2021.

CONFERENCE POSTERS

- 2026 **J. Yamamoto**[†], I. Bonamassa, and M. Pósfai. “When node weight matters: spectral transitions in physical networks.” Network Science Seminar 2025, Kyoto, Japan, February 17–20, 2026.
- 2022 **J. Yamamoto**[†], G. Shimojo, and K. Yakubo. “Bifractal nature of fractal scale-free networks and its implications.” Network Science Seminar 2022, Kyoto, Japan, August 23–25, 2022.
- 2021 **J. Yamamoto**[†] and K. Yakubo. “Bifractality of scale-free networks.” Network Science Seminar 2021, Kanazawa, Japan, December 11–12, 2021. *Poster Award*.

SEMINARS AND WORKSHOPS

- 2025 **J. Yamamoto**[†]. *Physical Networks as Network-of-Networks*. Network Science Study Group (ネットワーク科学勉強会), online. May 25, 2025.
- 2025 **J. Yamamoto**[†], I. Bonamassa, and M. Pósfai. *Reentrant localization in physical Laplacian eigenvectors*. DYNASNET Workshop on Graphs, Networks, and Brains, Budapest, Hungary. May 13, 2025.
- 2024 **J. Yamamoto**[†], I. Bonamassa, and M. Pósfai. *Laplacian Spectra and Angular Constraints in Physical Networks*. Annual DYNASNET Meeting, Valtice, Czech Republic. October 16, 2024.

FELLOWSHIPS, SCHOLARSHIPS, AND GRANTS

- 2026 AccelNet-MultiNet Fellowship, National Science Foundation. (\$4,000)
Research visit award at the Center for Complex Networks Research, Northeastern University.
- 2022–2023 ITO Scholarship for Japanese Students, ITO Foundation. (~\$50,000)
Tuition support, monthly stipend, and travel allowance for the MSc program at QMUL.
- 2019–2020 Scholarship for Study Abroad, Japan Student Services Organization. (~\$5,000)
Stipend for the exchange program at ETH Zürich.

AWARDS AND HONORS

2025	First-Year PhD Award, Central European University.
2023	Principal's Prize, Queen Mary University of London.
2022	Dean's Award for Academic Achievement, Hokkaido University.
2021	Poster Award, Network Science Seminar 2021, Kanazawa, Japan.
2019	Lane Memorial Award, Hokkaido University.
2018	Nitobe Award, Hokkaido University.

TEACHING

2024–2026	Teaching Assistant, Central European University. Scientific Python (graduate); Introduction to Linear Algebra (undergraduate).
2021–2023	Teaching Assistant, Hokkaido University. Statistical Mechanics I; Applied Mathematics II; Computational Science; Introduction to Statistics.

ACADEMIC SERVICE

2025–present	Ad hoc reviewer, <i>Nature Communications</i> ; <i>Journal of Complex Networks</i> ; <i>Network Science</i> ; <i>Transactions of the Japanese Society for Artificial Intelligence</i> .
2025–present	Organizer, Japanese Researcher Network in Austria.
2025–2026	Organizer, Physical Networks IV Satellite, NetSci2026, Boston, USA.
2024–2025	Organizer, Physical Networks III Satellite, NetSci2025, Maastricht, Netherlands.
2024–2025	Organizer, Data Stories 11th Edition, Department of Network and Data Science, Central European University.

MEMBERSHIPS

2026–present	American Physical Society.
2024–present	Austrian Physical Society.
2023–present	Network Science Society.
2020–present	Physical Society of Japan.

SKILLS

Languages	Japanese (native), English (fluent), German (basic).
Programming	Python, C++, Julia, Fortran.
Software	Linux, Git, L ^A T _E X, Mathematica.

REFEREES

Prof. Dr. Márton Pósfai, *PhD supervisor*. Department of Network and Data Science, Central European University, Vienna, Austria.

Prof. Dr. János Kertész, *PhD co-supervisor*. Department of Network and Data Science, Central European University, Vienna, Austria

Prof. Dr. Ginestra Bianconi, *MSc supervisor*. School of Mathematical Sciences, Queen Mary University of London, London, UK.

Prof. Dr. Kousuke Yakubo, *undergraduate supervisor*. National Institute of Technology, Asahikawa College, Asahikawa, Japan.

Prof. Dr. Ivan Bonamassa, *research collaborator*. International Research Center for Complexity Science, Hangzhou International Innovation Institute, Beihang University, Hangzhou, China.